

lab7

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A small retail store needs to assign three employees—Alice, Bob, and Charlie—to three distinct shifts: Morning (0), Afternoon (1), and Evening (2). Each employee must be assigned exactly one shift, and no two employees can work the same shift. Additionally, Alice is unavailable to work the Morning shift. Write a Python script using Google OR-Tools to model this Constraint Satisfaction Problem. Define the variables, apply the domains and constraints, and print the final consistent and complete assignment for all three employees.

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[ ]: print("some")
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Cryptarithmic is a classic logic puzzle where letters are replaced by numbers to form a valid mathematical equation. Write a Python script using Google OR-Tools to solve the equation: $\text{SEND} + \text{MORE} = \text{MONEY}$. You must define variables for the unique letters (S, E, N, D, M, O, R, Y) with a domain of 0 to 9. Apply constraints to ensure that every letter represents a strictly unique digit. Add constraints to prevent the leading letters ('S' and 'M') from being 0. Finally, create the linear constraint that mathematically represents the addition of the two words equaling the third. Print the integer value of each letter once the complete assignment is found.

A technology factory manufactures three products: Laptops, Phones, and Tablets. Each product requires a specific number of microchips, screens, and batteries. A Laptop requires 2 chips, 1 screen, and 1 battery, yielding a profit of \$500. A Phone requires 1 chip, 1 screen, and 1 battery, yielding a profit of \$300. A Tablet requires 1 chip, 1 screen, and 2 batteries, yielding a profit of \$400. The factory has a limited daily inventory of exactly 100 microchips, 80 screens, and 120 batteries. Furthermore, company policy dictates that at least 10 units of each product must be manufactured daily to meet vendor contracts. Write an OR-Tools script to formulate this Integer Linear Programming (ILP) problem. Apply the inventory and minimum production constraints, and define the objective function to maximize total daily profit. Output the maximum profit alongside the optimal quantity of Laptops, Phones, and Tablets to produce.